Computing set upper tolerance for the minimum spanning tree problem

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This is joint work with Gerold Jäger

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Combinatorial sum problems

Define combinatorial sum problem (CSP) $\mathcal{P} = (\mathcal{E}, D, c, f_c)$ as follows:

- $\mathcal{E}$: finite ground set of elements,
- D: set of feasible solutions,
- $c : \mathcal{E} \to \mathbb{R}$: cost function on $\mathcal{E}$,
- $f_c : D \to \mathbb{R}$: cost function on D with $f_c(S) = \sum_{e \in S} c(e)$ for $S \in D$.
- Let S^* be optimal, i.e., $f_c(S^*) = \min_{S \in D} f_c(S)$.
- Set $c^* := f_c(S^*)$.

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Famous examples of CSPs:

- Minimum spanning tree problem (MSTP),
- Traveling salesman problem,
- Shortest path problem.

Minimum spanning tree problem

Minimum spanning tree problem (MSTP):
problem of finding a spanning tree of minimum weight in a given weighted connected undirected graph.

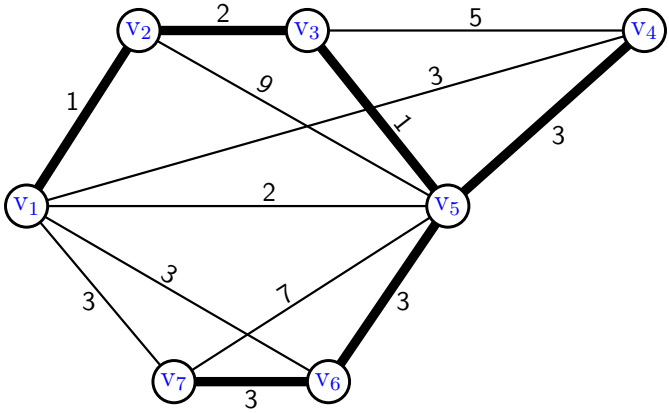
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Formally:
Let Γ be a connected undirected graph.
MSTP is a CSP $\mathcal{P} = (\mathcal{E}, D, c, f_c)$, where

- $\mathcal{E} = E(\Gamma)$,
- D is the set of spanning trees in Γ,
- $c : E(\Gamma) \to \mathbb{R}$,
- $f_c(T) = \sum_{e \in E(T)} c(e) \; \forall \; T \in D$.

Example



Single tolerances

Let $e \in \mathcal{E}$.

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Maximum by which the cost of e can be decreased such that the objective value of optimal solution remains the same.

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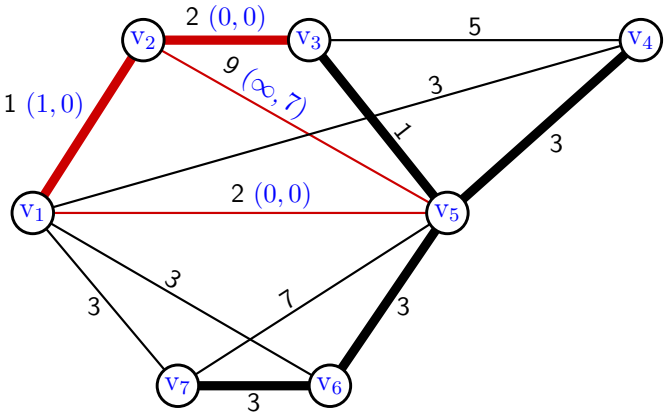
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Maximum by which the cost of e can be decreased such that the objective value of optimal solution remains the same.

Note that the costs of all other elements $\bar{e} \in \mathcal{E} \setminus \{e\}$ must remain unchanged.

Example revisited

Single upper and lower tolerances of individual edges



Set tolerances

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(Regular) set upper tolerance $u'_{\mathcal{P}}(E)$:
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Single tolerances for MSTP

- Recursive method for computing single upper tolerances for the MSTP in $\mathcal{O}(n^2)$ time, where n is the number of vertices.
  [Chin, Houck, 1978]
- Method for computing single lower tolerances for the MSTP with the same complexity.
  [Helsgaun, 2000]
- To our knowledge, only single tolerances have been considered for this problem, but not set tolerances.

F. Chin and D. Houck, Algorithms for updating minimal spanning trees, *Journal of Computer and System Sciences* **16**, 334–344 (1978).

K. Helsgaun, An effective implementation of the Lin-Kernighan traveling salesman heuristic, *European Journal of Operations Research* **126**(1), 106–130 (2000).

Set tolerances for MSTP

Proposition 1 [Jäger, P., 2026]

Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq E(\Gamma)$. Then $u'_{\mathcal{P}}(E) \geq \sum_{i=1}^{k} u'_{\mathcal{P}}(e_i)$.

Note that if E is a subset of edges of some minimum spanning tree, then $u'_{\mathcal{P}}(E) = \sum_{i=1}^{k} u'_{\mathcal{P}}(e_i)$ holds.

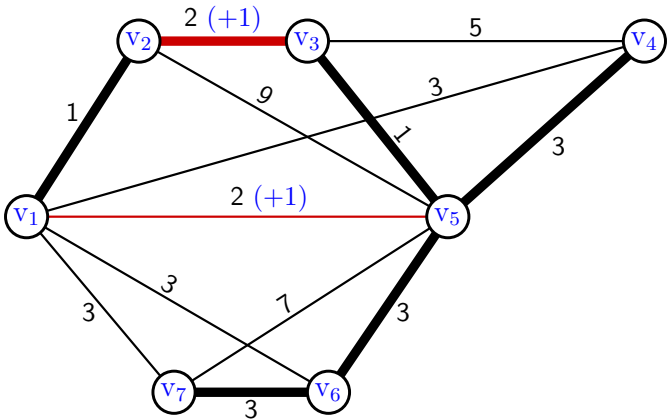
Proposition 2 [Jäger, P., 2026]

Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq E(\Gamma)$. Then $l'_{\mathcal{P}}(E) = \sum_{i=1}^{k} l'_{\mathcal{P}}(e_i)$.

G. Jäger and D. Panasenko, Computation of set tolerances with applications to the minimum spanning tree problem, *arXiv*, https://arxiv.org/abs/2605.10542 (2026).

Example revisited

Set tolerance for a set of two edges



Replacement edges

Let T be a spanning tree and $g \notin E(T)$. Denote by $P_T(g)$ the path in T joining the ends of g.

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Let e be an edge from $P_T(g)$. We will call g a replacement edge for e with respect to T, since removing e from T disconnects the tree into two parts and g connects these parts again.

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- Replacement set. The set $\mathrm{REP}(e)$ consists of e together with all replacement edges for e with respect to T^* that have the same cost as e, if such edges exist.

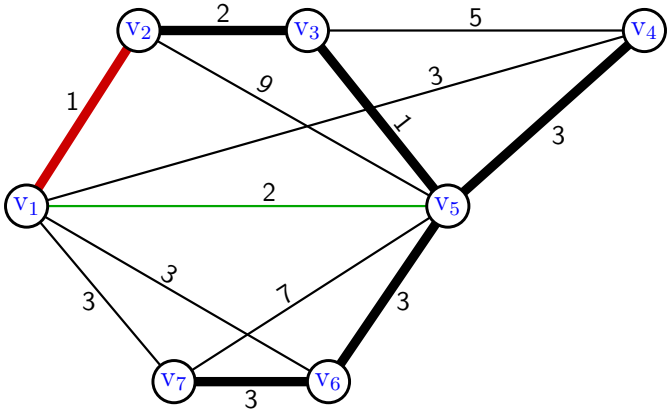
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- Replacement set. The set $\mathrm{REP}(e)$ consists of e together with all replacement edges for e with respect to T^* that have the same cost as e, if such edges exist.
- Increase of a replacement set. Let g be a replacement edge for e with respect to T^* with minimum cost among those satisfying $c(g) > c(e)$. The increase of $\mathrm{REP}(e)$ is defined as $\mathrm{inc}(\mathrm{REP}(e)) = c(g) - c(e)$. If such an edge g does not exist, then $\mathrm{inc}(\mathrm{REP}(e)) = \infty$.

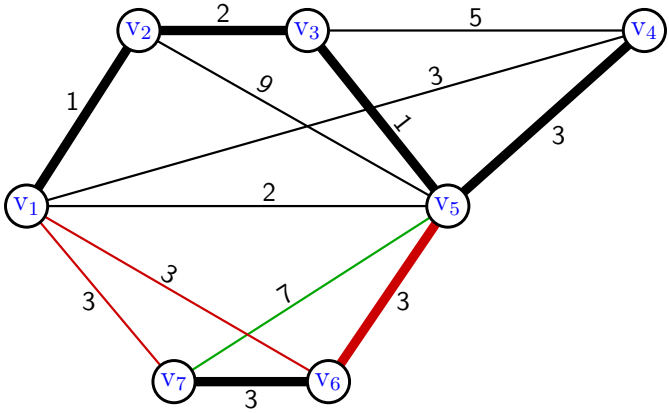
Example revisited

Replacement set with only one edge



Example revisited

Replacement set with several edges



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- Replacement class. Consider the graph whose vertices are the replacement sets, and where two vertices are adjacent if the corresponding sets have nonempty intersection. Then replacement class $C(e)$ is the union of all $\mathrm{REP}(\cdot)$ contained in the connected component of $\mathrm{REP}(e)$ in this graph.

Replacement classes

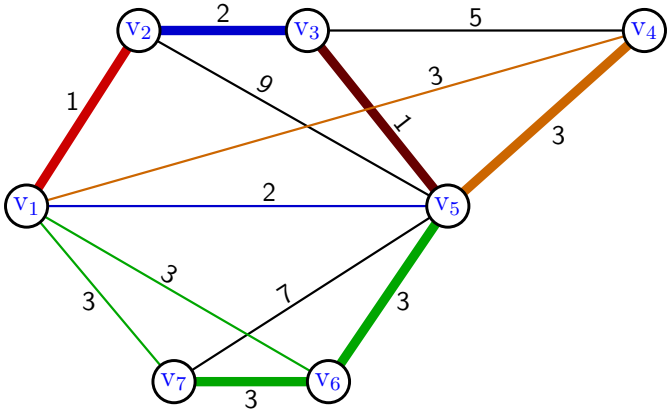
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- Increase of a replacement class. The increase of $C(e)$ is defined as

$$\mathrm{inc}(C(e)) = \min_{e' \in E(T^*) \cap C(e)} \{\mathrm{inc}(\mathrm{REP}(e'))\}.$$

Example revisited

All replacement classes



Properties of replacement classes

Define $U_{\mathcal{P}}$ as the set of all edges that belong to at least one MST of $\mathcal{P}$.

Lemma 1

Let $\mathcal{P} = (\mathcal{E}, D, c, f_c)$ be an MSTP and let T^* be an MST of $\mathcal{P}$. Then there exists a set of edges $\{e'_1, e'_2, \ldots, e'_l\} \subseteq E(T^*)$ such that the collection $\{C(e'_1), C(e'_2), \ldots, C(e'_l)\}$ forms a partition of the set $U_{\mathcal{P}}$.

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Lemma 2

Let $\{C(e'_1), C(e'_2), \ldots, C(e'_l)\}$ be the partition of the set $U_{\mathcal{P}}$ defined in Lemma 1. Then this partition is independent of the choice of T^*.

Properties of replacement classes

Lemma 3

Let $\mathcal{P} = (\mathcal{E}, D, c, f_c)$ be an MSTP and T', T'' be two distinct MSTs of $\mathcal{P}$. Also let $e' \in E(T')$ and $e'' \in E(T'')$ such that corresponding replacement classes $C'(e')$ and $C''(e'')$ are equal. Then $\text{inc}(C'(e')) = \text{inc}(C''(e''))$.

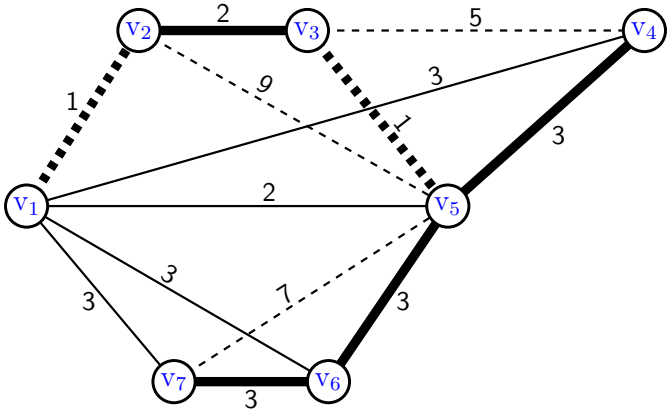
Let $\mathcal{P} = (\mathcal{E}, D, c, f_c)$ be an MSTP instance and $E = \{e_1, e_2, \ldots, e_k\} \subseteq \mathcal{E}$.

Denote increases in the cost of elements from E by $\vec{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_k)$.

Also denote by $\mathcal{P}_{\vec{\alpha},E}$ an instance obtained from $\mathcal{P}$ by increasing the cost of elements from E by $\vec{\alpha}$.

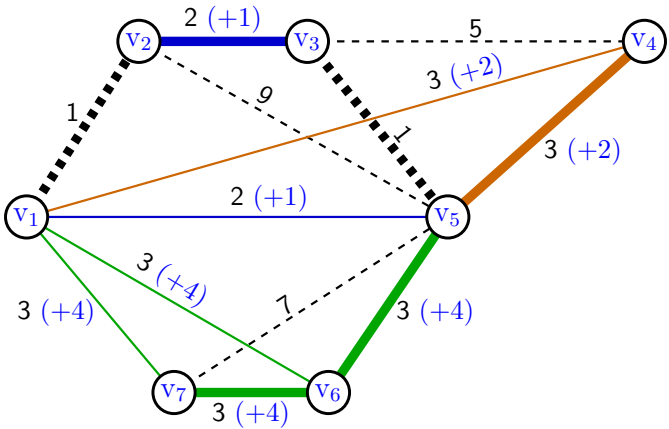
1: compute an MST T^* of $\mathcal{P}$
2: $u := 0$
3: **for** $j \in \{1, 2, \ldots, k\}$ **do**
4: $\quad \alpha_j := 0$
5: compute the set $U_{\mathcal{P}}$
6: **for** $j \in \{1, 2, \ldots, k\}$ **do**
7: $\quad$ **if** $e_j \notin U_{\mathcal{P}}$ **then**
8: $\quad\quad u := \infty; \alpha_j := \infty$
9: $\quad\quad$ **return** u, $\vec{\alpha}$
10: $\mathcal{P}' := \mathcal{P}_{\vec{\alpha},E}$
11: **for** $e \in E(T^*)$ **do**
12: $\quad$ compute $C'(e)$ and $\mathrm{inc}(C'(e))$
13: compute the set $U_{\mathcal{P}'} := \bigcup_{e \in E(T^*)} C'(e)$
14: compute the partition $\{C'(e'_1), C'(e'_2), \ldots, C'(e'_l)\}$ of $U_{\mathcal{P}'}$

15: $E' := E$
16: **for** $i \in \{1, 2, \ldots, l\}$ **do**
17: | **if** $C'(e'_i) \not\subseteq E$ **then**
18: | | $E' := E' \setminus C'(e'_i)$
19: **if** $E' == \emptyset$ **then**
20: | **return** u, $\vec{\alpha}$
21: **for** $i \in \{1, 2, \ldots, l\}$ **do**
22: | **if** $C'(e'_i) \subseteq E$ **then**
23: | | **for** $j \in \{1, 2, \ldots, k\}$ **do**
24: | | | **if** $e_j \in C'(e'_i)$ **then**
25: | | | | $u := u + \mathrm{inc}(C'(e'_i))$; $\alpha_j := \alpha_j + \mathrm{inc}(C'(e'_i))$
26: **if** $u == \infty$ **then**
27: | **return** u, $\vec{\alpha}$
28: **go to** 10


v2
v3
v4
v1
v5
v7
v6
2
5
1
9
3
1
3
2
3
3
7
3
3

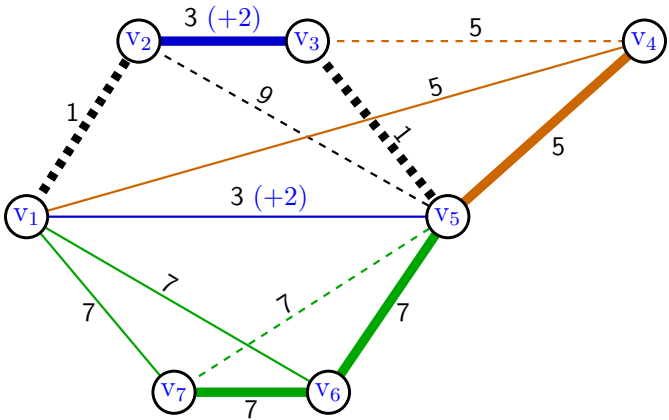
Example revisited

Iteration 1



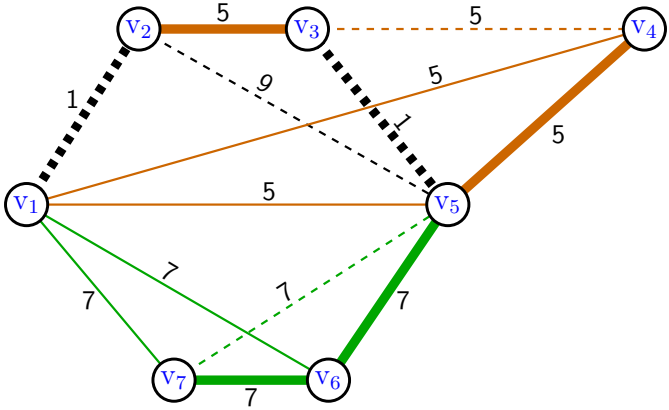
Example revisited

Iteration 2



Example revisited

Final iteration



$$u'_{\mathcal{P}}(E) = 4 \cdot 4 + 2 \cdot 2 + 2 \cdot 1 + 2 \cdot 2 = 26$$

Theorem 1

The value u produced by the proposed algorithm is equal to $u'_{\mathcal{P}}(E)$. The vector $\vec{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_k)$ produced by the proposed algorithm gives the increases for the elements of E, such that $\sum_{i=1}^{k} \alpha_i = u'_{\mathcal{P}}(E)$.

Correctness and complexity

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Theorem 2

The complexity of the proposed algorithm is $O(n^2 \cdot m)$, where m is the number of edges and n is the number of vertices of the instance.

Future work

- Improving complexity of the algorithm.
- Exploring ways to make the algorithm non-recursive.
- Exploring possible connections to matroid theory.

Thanks for your attention!